

Lab 10 - Gaussian Quadrature

Name: Lilith Janevski

Problem 1

```
f = @(x) x.^4;
h = 0;
CompArea = [];
TrueArea = 2/5;
a = -1;
b = 1;
xs = [];
Error = [];

i = 3;
I = ((b-a)/2).*sum(MyLegendreWeights(i).*f( ((b-a)/2).*MyLegendreRoots(i) +
((b+a)/2) ));
Error = [Error; abs(TrueArea-I)];
CompArea = [CompArea; I];

i = 5;
I = ((b-a)/2).*sum(MyLegendreWeights(i).*f( ((b-a)/2).*MyLegendreRoots(i) +
((b+a)/2) ));
Error = [Error; abs(TrueArea-I)];
CompArea = [CompArea; I];

i = 2;
xs = a:(b-a)/i:b;
I = MyTrapezoidQuad(f,xs,(b-a)/i);
Error = [Error; abs(TrueArea-I)];
CompArea = [CompArea; I];
I = MySimpsonQuad(f,xs,(b-a)/i);
Error = [Error; abs(TrueArea-I)];
CompArea = [CompArea; I];

i = 4;
xs = a:(b-a)/i:b;
I = MyTrapezoidQuad(f,xs,(b-a)/i) ;
Error = [Error; abs(TrueArea-I)];
CompArea = [CompArea; I];
I = MySimpsonQuad(f,xs,(b-a)/i);
Error = [Error; abs(TrueArea-I)];
CompArea = [CompArea; I];

n = [3; 5; 2; 2; 4; 4];
```

```

Legendre = [CompArea(1:2,1); 0; 0; 0; 0];
Trapezoidal = [0; 0; CompArea(3,1); 0; CompArea(5,1); 0];
Simpsons = [0; 0; 0; CompArea(4,1); 0; CompArea(6,1)]
t = table(n,Error,Legendre, Trapezoidal, Simpsons)

function [result] = MyTrapezoidQuad(f,xs,h)
    I = (h/2) * ((2*sum(f(xs))) - f(xs(1)) - f(xs(end)));
    result = I;
end

function [result] = MySimpsonQuad(f,xs,h)
    I = (h/3) * ((2*sum(f(xs))) + (2*sum(f(xs(2:2:end-1)))) - f(xs(1)) -
f(xs(end)));
    result = I;
end

function xs = MyLegendreRoots(n)
    syms x;
    xs = vpasolve(legendreP(n,x) == 0);
    xs = double(xs);
end

function ws = MyLegendreWeights(n)
    xs = MyLegendreRoots(n);
    syms x;
    derivative_L = diff(legendreP(n,x),x,1);
    ws = 2 ./ ((1 - xs.^2).*double(subs(derivative_L,x,xs).^2));
end

```

Ouput:

6×5 table

n	Error	Legendre	Trapezoidal	Simpsons
3	1.1102e-16	0.4	0	0
5	1.1102e-16	0.4	0	0
2	0.6	0	1	0
2	0.26667	0	0	0.66667
4	0.1625	0	0.5625	0
4	0.016667	0	0	0.41667

Well, nothing special about Trapezoidal and Simpson's that hasn't been said before. Simpson's has much faster convergence. However, Legendre's convergence is insanely fast even when compared to Simpson. Very impressive. Note that the error may just be a floating-point issue, since at this degree we should be able to get the exact value, as the degree of percision is $2n-1$ or less, and both of these fullfill that requirement for this 4th degree polynomial. This is backed up by the fact that the error doesn't even change.

Problem 2

(a)

6×5 table

n	Error	Legendre	Trapezoidal	Simpsons
3	3.9711e-05	0.28575	0	0
5	1.2586e-06	0.28572	0	0
2	0.052674	0	0.33839	0
2	0.0011965	0	0	0.28452
4	0.013077	0	0.29879	0
4	0.00012174	0	0	0.28559

(b)

6×5 table

n	Error	Legendre	Trapezoidal	Simpsons
3	1.8745e-06	1.9397	0	0
5	9.2322e-11	1.9397	0	0
2	0.018556	0	1.9212	0
2	0.00053548	0	0	1.9403
4	0.0046152	0	1.9351	0
4	3.1639e-05	0	0	1.9398

(c)

6×5 table

n	Error	Legendre	Trapezoidal	Simpsons
3	0.010402	1.1324	0	0
5	0.00029093	1.1217	0	0
2	0.088262	0	1.2102	0
2	0.0445	0	0	1.0775
4	0.0126	0	1.1346	0
4	0.012621	0	0	1.1093

(d)

6×5 table

n	Error	Legendre	Trapezoidal	Simpsons
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3	3.8002e-05	1.852	0	0
5	9.977e-07	1.8519	0	0
2	0.06654	0	1.7854	0
2	0.004994	0	0	1.8569
4	0.01643	0	1.8355	0
4	0.00027339	0	0	1.8522

In a, we immediately see Legendre struggle as this is no longer a polynomial, it's still the best but doesn't perform as well as it did in the prior integral. b progresses as normal, but since the interval is so short we don't get the benefit of periodicity to speed up the convergence. c seems normal. d being periodic but decreasing means we don't benefit from periodicity, and everything proceeds as normal with Legendre being the fastest convergence, Simpson's next best, and Trapezoidal the worst.

Problem 3

```
f = @(x) (cos(x))./sqrt(1-x.^2);
CompArea = [];
TrueArea = 2.4039394306344129;
a = -1;
b = 1;
Error = [];

i = 3;
I = ((b-a)/2).*sum(MyLegendreWeights(i).*f( ((b-a)/2).*MyLegendreRoots(i) +
((b+a)/2) ));
Error = [Error; abs(TrueArea-I)];
CompArea = [CompArea; I];

I = sum((pi/i).*cos(MyChebyshevRoots(i)));
Error = [Error; abs(TrueArea-I)];
CompArea = [CompArea; I];

i = 5;
I = ((b-a)/2).*sum(MyLegendreWeights(i).*f( ((b-a)/2).*MyLegendreRoots(i) +
((b+a)/2) ));
Error = [Error; abs(TrueArea-I)];
CompArea = [CompArea; I];

I = sum((pi/i).*cos(MyChebyshevRoots(i)))
Error = [Error; abs(TrueArea-I)];
CompArea = [CompArea; I];

n = [3; 3; 5; 5];
Legendre = [CompArea(1,1); 0; CompArea(3,1); 0];
Chebyshev = [0; CompArea(2,1); 0; CompArea(4,1)];
```

```

t = table(n,Error,Legendre, Chebyshev)

function xs = MyLegendreRoots(n)
    syms x;
    xs = vpasolve(legendreP(n,x) == 0);
    xs = double(xs);
end

function ws = MyLegendreWeights(n)
    xs = MyLegendreRoots(n);
    syms x;
    derivative_L = diff(legendreP(n,x),x,1);
    ws = 2 ./ ((1 - xs.^2).*double(subs(derivative_L,x,xs).^2));
end

function xs = MyChebyshevRoots(n)
    points = 2 .* (1:n) - 1;
    xs = cos(points .* pi ./ (2*n));
end

```

Output:

n	Error	Legendre	Chebyshev
3	0.25945	2.1445	0
3	0.00013156	0	2.4041
5	0.16871	2.2352	0
5	1.6529e-09	0	2.4039

Wow, the method that works only for hyper-specific functions works really well for those hyper-specific functions, who'd've thought!? Is there much more to say than Chebyshev being superior?